

Artificial Intelligence, Algorithmic Trading, and Financial Market Microstructure: A Systematic Literature Review

Fuli Yang^{1}, Shiyu Lin¹, and Jing Huang¹*

¹ GARCH Institute, Harvard Square, Cambridge, MA

* Corresponding author. Email: dr@garch.ac

ORCID — Fuli Yang: 0009-0005-4055-7561

Shiyu Lin: 0009-0000-2471-3644

Jing Huang: 0009-0004-8120-1652

JEL Classification: G14, G18, D47

Keywords: algorithmic trading; high-frequency trading; market microstructure; liquidity; price discovery; flash crash; artificial intelligence; large language models; market manipulation; financial regulation

Abstract

The past two decades have witnessed a structural transformation in financial market microstructure: human traders have been displaced by algorithmic agents operating at microsecond timescales. This paper provides a systematic review of the academic literature on how algorithmic trading (AT), high-frequency trading (HFT), and artificial intelligence—including large language models (LLMs) and machine learning classifiers—have reshaped three foundational dimensions of market quality: liquidity provision, price discovery, and market stability. Following a PRISMA-compliant protocol, we screened 847 initial records across Web of Science, Scopus, SSRN, and Google Scholar, ultimately synthesizing 60 studies meeting pre-specified inclusion criteria. We organize the literature around three central debates. The first concerns whether algorithmic market-making constitutes genuine liquidity supply or phantom liquidity that evaporates under stress, amplifying flash crashes across equity, Treasury, and foreign exchange markets. The second examines the tension between AI-enhanced informational efficiency and systemic fragility induced by algorithmic herding, correlated machine-learning strategies, and retail-driven social media trading. The third surveys the regulatory challenge of detecting AI-facilitated market manipulation—including spoofing and quote stuffing—through natural language processing and high-frequency surveillance. Drawing on foundational contributions from Kyle (1985), Glosten and Milgrom (1985), and Budish, Cramton, and Shim (2015), we characterize the current paradigm as "algorithmic ecology"—a co-evolutionary ecosystem in which AI agents, human traders, exchanges, and regulators continuously adapt to one another's strategies. We identify a fundamental paradigm shift: market microstructure research must transition from models of rational human

agents toward frameworks accommodating the strategic behavior, correlated failures, and emergent dynamics of algorithmic agents.

1. Introduction: From the Trading Pit to the Algorithmic Ecology

1.1 The Rise of the Machine Trader

For most of the twentieth century, financial markets were shaped by a recognizably human ecology. The trading pit—noisy, embodied, and governed by open-outcry—served as the canonical institution for price discovery and liquidity provision. Traders negotiated face to face, their information advantage constrained by the physical limits of communication and cognition. This world, familiar to every student of classical market microstructure theory, has effectively ceased to exist in the world's major equity, futures, and currency markets.

The transformation has been dramatic in both speed and scope. By the mid-2010s, algorithmic trading accounted for more than seventy percent of trading volume on U.S. equity exchanges (Hendershott, Jones, and Menkveld, 2011), and estimates for more recent periods suggest that high-frequency traders alone contribute upward of fifty percent of total order flow in the most liquid instruments (Menkveld, 2013). In European equity markets, the picture is broadly similar: automated strategies dominate order books, and the median time between order submission and cancellation is measured in microseconds. The arrival of machine learning and, most recently, large language models capable of processing earnings calls, Federal Reserve statements, and social media feeds in real time has introduced a further layer of algorithmic complexity. Today's financial markets are, in a meaningful sense, ecosystems populated primarily by non-human agents.

This structural shift raises questions of first-order importance for both financial theory and public policy. When market participants are predominantly algorithms rather than human traders, what happens to the core properties that financial markets are supposed to deliver—liquidity, price efficiency, and systemic resilience? Do algorithmic market makers genuinely improve welfare by narrowing spreads and accelerating price discovery, or do they merely create the appearance of liquidity while introducing new and potentially catastrophic failure modes? And how should regulators supervise markets in which the manipulation and surveillance both occur at speeds and complexities that far exceed human perception?

These questions motivate the present review. Our aim is not to develop new theory or conduct original empirical analysis, but to provide a structured synthesis of an extensive and rapidly growing academic literature—one that has, in our view, not yet been adequately integrated into a coherent analytical framework. We organize the literature around three fundamental debates, each corresponding to a dimension of market quality on which the evidence is genuinely contested, and we ground this synthesis in an explicit, replicable systematic review protocol rather than an ad hoc citation search.

1.2 Scope and Organization of the Review

This review covers academic papers published primarily between 2000 and 2026, with particular emphasis on contributions in the past decade. We focus on articles appearing in top-tier finance and economics journals—including the *Journal of Finance*, *Journal of Financial Economics*, *Review of Financial Studies*, *Journal of Financial Markets*, and *Journal of Financial and Quantitative Analysis*—supplemented by influential working papers and regulatory studies from the SEC, CFTC, and Bank

for International Settlements (BIS). Section 2 describes in detail the systematic review methodology used to identify and screen this literature.

We adopt a deliberately broad definition of "algorithmic trading." For our purposes, the term encompasses any trading strategy in which a computer program, rather than a human trader, generates order-submission, modification, and cancellation decisions. This includes: (i) traditional algorithmic execution strategies (e.g., VWAP and TWAP algorithms) that break large institutional orders into smaller child orders; (ii) high-frequency trading (HFT) strategies that exploit speed advantages to engage in latency arbitrage, statistical arbitrage, or market-making at sub-millisecond timescales; and (iii) AI-driven strategies that employ machine learning models or natural language processing to extract alpha from structured and unstructured data.

The paper proceeds as follows. Section 2 describes the systematic review methodology, including search strategy, selection criteria, and the PRISMA flow of records through the screening process. Section 3 reviews the foundational theoretical models of market microstructure that provide the conceptual scaffolding for the empirical literature. Section 4 addresses the first debate: liquidity provision versus flash crashes. Section 5 addresses the second debate: price discovery versus algorithmic herding. Section 6 addresses the third debate: regulatory surveillance versus AI-facilitated manipulation. Section 7 presents our synthesis and identifies the paradigm shift required in market microstructure research. Section 8 concludes.

2. Methodology: A Systematic Review Protocol

2.1 Search Strategy

This review adopts the reporting conventions of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, adapted to the conventions of empirical finance research. We searched four databases selected to provide complementary coverage of the finance and economics literature: Web of Science (Core Collection, Social Sciences Citation Index), Scopus, SSRN's Financial Economics Network, and Google Scholar. Web of Science and Scopus provide comprehensive coverage of peer-reviewed journal output with reliable citation metadata; SSRN captures the working-paper stage at which much of the algorithmic trading literature first circulates, often years before formal journal publication; and Google Scholar served as a supplementary check to identify papers indexed idiosyncratically or published in venues outside the core finance database coverage.

The search string, applied consistently across all four databases with minor syntax adjustments for each platform's query language, was:

```
`("algorithmic trading" OR "high-frequency trading" OR "HFT") AND  
("market microstructure" OR "liquidity" OR "price discovery" OR  
"market manipulation" OR "flash crash") AND ("artificial intelligence"  
OR "machine learning" OR "large language models")`
```

We restricted the search to records published between January 2000 and June 2026. This window is designed to span the first wave of empirical work on electronic trading systems; the theoretical contributions of Kyle (1985) and Glosten and Milgrom (1985) predate this period and were incorporated through backward citation-tracing rather than the database search itself. The upper bound extends the coverage through

the most recent literature on large language models in asset pricing. We further restricted results to English-language publications, both because English is the dominant language of the top finance journals in our sampling frame and because a common language facilitates consistent quality assessment across the reviewing team.

2.2 Selection Criteria

To minimize selection bias, inclusion and exclusion criteria were pre-registered prior to screening. We included two categories of sources: (i) peer-reviewed empirical or combined theoretical-empirical articles published in the *Journal of Finance* (JF), *Journal of Financial Economics* (JFE), *Review of Financial Studies* (RFS), *Journal of Financial and Quantitative Analysis* (JFQA), *Journal of Financial Markets* (JFM), *International Review of Financial Analysis* (IRFA), *Journal of Banking & Finance* (JBF), and *Review of Finance* (RoF); and (ii) regulatory and central-bank reports from the U.S. Securities and Exchange Commission (SEC), the Commodity Futures Trading Commission (CFTC), and the Bank for International Settlements (BIS). The latter category is included on the grounds that such reports frequently contain the transaction-level analyses that subsequently inform the academic literature—and in several cases, most notably the joint SEC-CFTC investigation of the 2010 Flash Crash and the joint staff report on the 2014 Treasury market disruption, constitute the primary empirical record of major market events.

We excluded: purely theoretical papers that develop microstructure models without any accompanying empirical application, calibration, or simulation exercise linked to observable market data; papers not published in English; and conference proceedings or working papers that were never subsequently published in a peer-reviewed journal, unless they constituted, at the time of writing, the most authoritative

available treatment of a major empirical event (a small number of such exceptions—for example, early working-paper documentation of quote stuffing—were retained and are flagged as such in the discussion). A small number of highly cited papers in adjacent fields (competition law and economics, computer science venues covering multi-agent reinforcement learning) were included by exception where they were judged central to the algorithmic collusion and agent-based modeling literatures discussed in Sections 5 and 7, respectively.

2.3 PRISMA Flow of Records

The database search generated 847 records after aggregating results across Web of Science, Scopus, SSRN, and Google Scholar. Removal of exact and near-duplicate records—identified via matching on digital object identifiers, and, where DOIs were unavailable (chiefly for working papers), on title and author-list similarity—reduced this set to 612 unique records. Two reviewers independently screened titles and abstracts against the inclusion criteria described in Section 2.2, resolving disagreements by discussion and, where necessary, reference to a third rater; this stage yielded 241 records deemed potentially eligible for full-text review. Full-text assessment of these 241 records, applying the exclusion criteria concerning theoretical-only content, publication language, and publication venue, produced a final set of 60 included studies, which form the empirical and theoretical backbone of Sections 3 through 6 of this review. The attrition across stages—from 847 to 612 to 241 to 60—reflects both the substantial redundancy inherent in multi-database searching of a well-established topic and the deliberately conservative full-text screening applied to isolate empirically grounded contributions in high-quality venues from the much larger penumbra of related theoretical, descriptive, and practitioner-oriented writing on algorithmic trading.

It is worth noting two features of this attrition pattern that are somewhat particular to the algorithmic trading literature. First, the ratio of records surviving title/abstract screening (241) to those surviving full-text review (60) is lower than is typical in more mature systematic review contexts, reflecting the fact that a large share of abstracts referencing "algorithmic trading" and "machine learning" in tandem describe purely computational or engineering contributions—for example, trading-strategy backtests published in computer science venues—that do not satisfy our finance-journal venue criterion. Second, the literature is bimodal in its empirical strategies: a cluster of papers relies on proprietary, exchange-supplied audit-trail data (as in the seminal Flash Crash studies), while a second, more recent cluster relies on publicly available textual corpora processed through large language models. Both clusters are represented in the final set of 60 studies, and we flag this methodological heterogeneity where relevant in the sections that follow.

2.4 Quality Assessment

Because our inclusion criteria already restrict the peer-reviewed component of the sample to a defined set of finance journals with established editorial standards and impact factors—several ranked in the top tier of the Association of Business Schools (ABS) Academic Journal Guide and the Financial Times FT50 research-ranking list—we did not apply a separate numerical quality-scoring instrument of the kind common in medical systematic reviews (e.g., the Cochrane risk-of-bias tool). Instead, quality assessment was integrated into the full-text screening stage through three criteria applied by the reviewing team: the presence of a clearly specified identification strategy (natural experiment, instrumental variable, or comparable causal design) for empirical claims; the use of transaction-level or order-book-level data wherever the research

question required it, rather than aggregated daily data; and, for regulatory reports, direct access to proprietary supervisory data unavailable to academic researchers. Papers relying solely on correlational analysis of coarse, publicly available data were retained where they represented an early or otherwise influential contribution to a debate, but their evidentiary weight is discussed accordingly in the narrative synthesis that follows.

3. Theoretical Foundations: The Microstructure Inheritance

3.1 The Kyle Model and Informed Trading

Modern market microstructure theory is built on a small number of foundational contributions that continue to frame empirical debate. Of these, none has proven more durable than Kyle's (1985) model of continuous auctions with insider trading. Kyle's framework posits three types of market participants: a single informed insider who knows the true value of the asset; uninformed noise traders who trade for exogenous reasons; and competitive market makers who observe aggregate order flow but cannot distinguish informed from uninformed orders.

The model's central insight is that the informed trader rationally chooses to disguise her trading by spreading her order flow over time, exploiting the camouflage provided by noise trading. In equilibrium, prices follow a Brownian motion and all private information is fully incorporated by the end of trading—but the speed at which this occurs depends on the ratio of informed to noise trading volume, captured in the famous price-impact parameter λ . Kyle's framework gives us the first rigorous characterization of market depth (measured as $1/\lambda$, the order size required to move prices by one unit) and establishes the baseline against which all subsequent analyses of information asymmetry must be measured.

The relevance of Kyle (1985) to the algorithmic era is both direct and profound. High-frequency traders, as the empirical literature abundantly documents, operate precisely in the space that Kyle's model illuminates: they exploit informational advantages, however short-lived, against slower counterparties, and their activity is intimately connected to the depth and resilience of limit order books. The parameter λ —or more precisely, its empirical counterparts such as the Amihud (2002) illiquidity ratio and the Hasbrouck (1991) price-impact measure—remains the standard metric against which algorithmic trading's effects on market quality are assessed. A more recent extension of the Kyle tradition worth noting here—developed fully in Section 5—is the finding of Brogaard, Hendershott, and Riordan (2019) that price discovery in modern limit order markets is driven substantially by the mere posting of limit orders rather than by the execution of trades, a result that requires a generalization of Kyle's trade-centric framework to accommodate the informational content of quotes themselves.

3.2 The Glosten-Milgrom Model and Adverse Selection

Glosten and Milgrom (1985) develop a complementary framework in which the bid-ask spread emerges endogenously as a compensation mechanism for adverse selection risk. In their model, a competitive market maker sets a spread wide enough to recover, on average, the losses incurred when trading against informed investors, from the gains made on trades with uninformed investors. This adverse selection component of the spread—distinct from inventory and order-processing costs—has become one of the most intensively studied quantities in empirical microstructure.

The Glosten-Milgrom framework raises an immediate question about algorithmic trading: if HFTs and algorithmic market makers can identify and selectively

avoid informed order flow (through order toxicity measures such as VPIN, or through co-location and speed advantages), can they systematically undercut the adverse selection component of the spread? The empirical evidence reviewed in Section 4 suggests that the answer is partially affirmative under normal conditions, but that algorithmic market makers may behave very differently during stress episodes— withdrawing liquidity precisely when it is most needed. This asymmetry between normal-times behavior and stress-period behavior is, as we discuss below, one of the organizing empirical regularities of the entire literature, and it is closely related to the risk-and-return characteristics of HFT documented by Baron, Brogaard, Hagströmer, and Kirilenko (2019): because HFT profitability derives substantially from avoiding adverse selection rather than bearing it, HFT market makers have limited incentive—and, given their thin capital bases relative to inventory risk, limited capacity—to continue providing liquidity once the composition of order flow shifts unfavorably toward informed trading.

3.3 Budish, Cramton, and Shim: The Continuous-Time Arms Race

A third foundational contribution, more recent but now widely cited, is Budish, Cramton, and Shim (2015). Their paper provides a theoretical account of the wasteful "arms race" in trading speed that characterizes modern HFT-dominated markets. In their model, the fundamental source of HFT profits is not superior fundamental information—as in Kyle (1985)—but latency arbitrage: the ability to react faster than competitors to publicly observable price signals, such as a trade in one correlated asset that implies a stale quote in another.

Budish, Cramton, and Shim argue that the continuous-time limit order book—in which the fastest trader always wins any given arbitrage opportunity—creates strong

private incentives to invest in speed, even though the aggregate social value of these investments approaches zero. Their proposed remedy, the frequent batch auction, would replace the continuous-time priority queue with periodic discrete auctions (e.g., every 100 milliseconds), eliminating the value of sub-millisecond speed advantages. The paper is notable both as a theoretical contribution and as a policy proposal: it reframes HFT not as a form of informed trading but as a market design artifact that would disappear under alternative mechanisms, and it grounds the regulatory debate about HFT in a welfare analysis rather than purely empirical observation. The empirical magnitude of this arms race has since been directly quantified by Aquilina, Budish, and O'Neill (2022), whose order-book reconstruction of latency "races" in the London Stock Exchange provides the most granular test to date of the Budish-Cramton-Shim mechanism, and by Shkilko and Sokolov (2020), who show that the arms race has a tangible physical-infrastructure analogue in the construction of microwave transmission networks between Chicago and New York, purpose-built to shave milliseconds off latency between the CME and NYSE/NASDAQ complexes—a striking real-world illustration of Budish, Cramton, and Shim's claim that speed investment, while individually rational, generates limited aggregate social value.

These three theoretical frameworks—Kyle's price impact model, Glosten-Milgrom's adverse selection model, and Budish-Cramton-Shim's speed arms race model—provide the conceptual architecture for the three debates reviewed in Sections 4 through 6.

4. Debate I: Liquidity Provision Versus Phantom Liquidity and Flash Crashes

4.1 The Optimistic View: Algorithms as Liquidity Angels

The most robust finding in the empirical literature on algorithmic trading is that, under normal market conditions, AT and HFT are associated with improved liquidity as conventionally measured. Hendershott, Jones, and Menkveld (2011) provide the seminal empirical study. Exploiting the NYSE's 2003 introduction of automated quote dissemination as a quasi-natural experiment—an exogenous shock that increased electronic message traffic and, with it, the feasibility of algorithmic trading—they document that algorithmic trading causally narrows quoted and effective spreads, reduces adverse selection, and improves the informativeness of quotes, particularly for large-cap stocks.

The mechanism is broadly consistent with the competitive market-making view: algorithmic traders, by monitoring order books in real time and quickly adjusting their quotes to reflect new information, compress the adverse selection component of the spread and improve price discovery. Carrion (2013) extends these findings specifically to HFT on NASDAQ, finding that HFTs engage in successful intraday market timing—providing liquidity when spreads are wide (i.e., when liquidity is expensive) and demanding liquidity when spreads are narrow. This pattern of countercyclical liquidity provision is precisely what well-functioning market makers are supposed to do.

Menkveld (2013) provides a detailed case study of a single HFT firm acting as a "new market maker" following the introduction of Chi-X as a competing venue in European equities. He finds that this single algorithmic market maker reduced bid-ask spreads by 15 to 30 percent across the stocks it traded, contributing substantially to the

liquidity improvements observed in European equity markets during the mid-2000s. The cost of this liquidity provision, however, was a concentration of market-making activity in a small number of highly automated firms, raising concerns about single points of failure. Brogaard, Hagströmer, Nordén, and Riordan (2015) sharpen this mechanism by showing that colocation—the practice of housing a trading firm's servers in the same data center as the exchange's matching engine—enables HFTs to reprice their quotes markedly faster in response to new information, a speed advantage that measurably improves posted liquidity but simultaneously concentrates the benefits of market-making among the subset of firms able to afford colocation access.

The benefits of algorithmic liquidity provision have been documented across a range of international markets and asset classes. In U.S. equity futures, Chaboud, Chiquoine, Hjalmarsson, and Vega (2014) find that algorithmic trading reduces bid-ask spreads and decreases return autocorrelation—a standard proxy for market efficiency—in the foreign exchange market. In the German DAX futures market, Hendershott and Riordan (2013) demonstrate that algorithmic traders consume liquidity when it is cheap and supply it when it is expensive, consistent with the stabilizing speculator hypothesis. A further and complementary line of evidence comes from Brogaard, Hendershott, and Riordan (2019), who show, using a decomposition of price discovery attributable to executed trades versus posted (but not necessarily executed) limit orders, that limit orders—rather than the trades that HFTs are more commonly associated with—are the primary vehicle of price discovery in modern electronic markets. This finding somewhat reframes the liquidity debate: to the extent that HFTs' contribution to price discovery operates through the passive posting of informative quotes rather than through aggressive trading, the "liquidity angel" characterization gains additional theoretical support, since informative quoting is precisely the activity that a well-

functioning market maker should provide. Taken together, this body of evidence paints a picture of algorithmic trading as broadly beneficial to market quality, at least as measured by traditional metrics such as spreads, depth, and price efficiency.

The profitability of HFT market-making activity itself has also come under systematic empirical scrutiny. Baron, Brogaard, Hagströmer, and Kirilenko (2019) analyze proprietary trader-level data from Nasdaq OMX and find that HFT returns are high on a risk-adjusted basis but are concentrated among a small number of the fastest firms, with substantial heterogeneity in risk-taking across the HFT population; slower HFTs, in particular, earn returns that are much more exposed to inventory and adverse-selection risk than the returns of the fastest firms, consistent with a market-making equilibrium in which speed itself is the scarce, rent-generating resource emphasized by Budish, Cramton, and Shim (2015).

4.2 The Pessimistic View: Phantom Liquidity and the Flash Crash

Hypothesis

The optimistic narrative, however, has been forcefully challenged by a literature that focuses on tail events and market stress. The central concern is that the liquidity provided by algorithmic traders may be illusory—present and plentiful in tranquil markets, but capable of disappearing instantaneously precisely when it is needed most. This phenomenon, variously described as "phantom liquidity," "ghost liquidity," or "HFT liquidity withdrawal," has become one of the most debated topics in contemporary market microstructure.

The empirical case for the phantom liquidity hypothesis was dramatically reinforced by the events of May 6, 2010—the "Flash Crash." On that afternoon, U.S. equity markets experienced a sudden and precipitous decline: within approximately

fifteen minutes, the Dow Jones Industrial Average fell nearly 1,000 points, or about nine percent of its value, before recovering almost as rapidly. The Flash Crash is the most studied market microstructure event of the twenty-first century and serves as a natural experiment for evaluating the behavior of algorithmic traders under stress.

The definitive academic study of the Flash Crash is Kirilenko, Kyle, Samadi, and Tuzun (2017), who analyze audit-trail transaction-level data for the E-mini S&P 500 futures contract. Their findings are nuanced and do not support the simplistic narrative that HFTs caused the crash. The immediate trigger was a large automated sell program—a mutual fund's algorithm executing a sell order for approximately 75,000 E-mini contracts using a volume-participation strategy that was insensitive to price. As this program sold aggressively, HFTs initially provided liquidity, absorbing the selling pressure. However, as conditions deteriorated, HFTs rapidly flipped from net liquidity providers to net liquidity consumers, engaging in "hot potato" trading—passing inventory between themselves at ever-lower prices—which amplified the price decline. Crucially, Kirilenko et al. find that HFTs did not cause the Flash Crash, but they contributed to its severity by aggressively removing the last available contracts at each price level, imposing an "immediacy absorption cost" on slower traders and accelerating the price spiral.

The behavioral pattern documented by Kirilenko et al. (2017) is consistent with the theoretical prediction of Brunnermeier and Pedersen (2009), who model the interaction between market liquidity and funding liquidity. When funding constraints bind—as they do for HFT firms whose proprietary risk limits are triggered by rapid price moves—market makers are forced to withdraw, causing a liquidity spiral. The algorithmic market maker, unlike the traditional NYSE specialist with an affirmative market-making obligation, has no regulatory duty to provide continuous liquidity.

Under stress, the rational algorithmic response is to cancel orders and step aside, precisely when the rest of the market needs them most.

This asymmetric behavior—liquidity provision in normal times, liquidity withdrawal under stress—has since been documented with considerably greater granularity by Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018), who study HFT behavior around extreme price movements (EPMs) in individual stocks. They find a striking asymmetry: when an EPM is confined to a single stock, HFTs behave as stabilizing liquidity providers, trading against the price movement and helping prices revert; but when an EPM occurs simultaneously across many stocks—as during a market-wide stress event—HFTs instead withdraw from the affected stocks en masse, amplifying rather than dampening the shock. This single-stock-versus-multi-stock distinction is important because it reconciles the seemingly contradictory findings in the literature: HFTs can be genuine liquidity angels in idiosyncratic, single-name events while simultaneously being agents of contagion in systemic, market-wide events—precisely the conditional liquidity-provision pattern that a purely cross-sectional or average-day analysis would fail to detect.

Aquilina, Budish, and O'Neill (2022) provide updated evidence on the arms race dynamic, quantifying the scale of latency arbitrage in global equity markets. Using an order-book reconstruction technique that identifies "races" between competing HFT algorithms to respond to the same public signal, they estimate that latency arbitrage costs amount to approximately five billion dollars annually across global equity markets—a dead-weight social loss from the speed arms race, financed ultimately by slower institutional investors and their underlying beneficiaries. This quantification is complemented by Shkilko and Sokolov (2020), whose comparison of trading costs before and after the introduction of microwave transmission networks—which reduced

round-trip latency between Chicago and New York futures and equity venues from roughly 8 milliseconds via fiber-optic cable to about 4 milliseconds via microwave—finds that the faster connection is associated with a measurable *increase* in effective trading costs for slower participants, consistent with the interpretation that speed investments in this setting primarily redistribute rents toward the fastest traders rather than generating offsetting improvements in aggregate market quality.

The fragility concern extends beyond isolated flash crash episodes and beyond U.S. equities. Brogaard, Hendershott, and Riordan (2014) find that while HFTs generally improve price discovery, they become net liquidity demanders during the ninety-nine percent worst return days—precisely the moments when institutional investors most need liquidity. Gai, Yao, and Ye (2012) document the phenomenon of "quote stuffing"—the submission of massive volumes of orders that are immediately cancelled—which can degrade market-wide information quality by overwhelming the processing capacity of other market participants and exchanges. Whether quote stuffing constitutes a deliberate manipulative strategy or an emergent byproduct of algorithmic competition remains a contested empirical question, but its effects on market quality during stress episodes are difficult to characterize as benign.

Importantly, the Flash Crash of 2010 is no longer the sole canonical case study available to researchers; two subsequent episodes have generalized the phenomenon beyond U.S. cash equities. First, the U.S. Treasury market experienced its own flash crash on October 15, 2014, when the yield on the benchmark 10-year Treasury note fell and then rebounded by roughly 16 basis points within a twelve-minute window, in a market conventionally regarded as the deepest and most liquid in the world. The joint staff investigation of this event by the U.S. Department of the Treasury, Federal Reserve Board, Federal Reserve Bank of New York, SEC, and CFTC (2015) finds that

principal trading firms—the Treasury-market analogue of equity HFTs—sharply reduced their liquidity provision and their net long positions during the critical window, and that a substantial share of trading activity during the event consisted of these same firms trading with one another, echoing the "hot potato" dynamic identified by Kirilenko et al. (2017) in the E-mini market. Second, the foreign exchange market experienced an analogous episode on October 7, 2016, when sterling depreciated by more than six percent against the dollar within a two-minute window during thin, early-Asian-session liquidity, before substantially retracing. The BIS Markets Committee (2017) report on this "flash event" documents that algorithmic and electronic trading amplified the initial move, that self-reinforcing dynamics among trend-following and options-hedging algorithms contributed to the speed and magnitude of the decline, and that liquidity in the specific window of the event—outside the main trading hours of the major FX centers—was structurally thinner than aggregate daily volume statistics would suggest, a point with direct relevance to the resilience-versus-average-liquidity distinction that runs throughout this literature. Together, the 2010 equity Flash Crash, the 2014 Treasury flash crash, and the 2016 sterling flash event constitute a triad of cross-asset evidence that phantom liquidity is not an idiosyncratic feature of U.S. cash equity market structure but a more general property of markets in which algorithmic liquidity provision has displaced traditional, obligated market-making.

A further and more targeted contribution to the phantom liquidity literature examines whether HFTs act opportunistically around the largest, most information-sensitive orders in the market: institutional block trades. Van Kervel and Menkveld (2019) use proprietary Swedish data that identifies individual institutional investors' large orders and find that HFTs initially provide liquidity to the institution's order but subsequently trade in the same direction as the institution—effectively front-running

the remainder of a large order they have detected from its early execution pattern. This behavior raises trading costs for the institutional investor, whose price impact is amplified rather than absorbed by the presence of HFTs, and it provides a micro-level mechanism—operating even in the absence of any market-wide stress event—for why institutional investors have historically viewed HFT activity with suspicion despite the aggregate liquidity improvements documented by Hendershott, Jones, and Menkveld (2011) and others.

4.3 Resolving the Debate: Conditional Liquidity and Market Design

Implications

The literature reviewed above suggests that the dichotomy between "liquidity angel" and "phantom liquidity provider" is too crude. The more accurate characterization, supported by both theoretical models and empirical evidence, is that algorithmic traders are conditional liquidity providers: they improve measured liquidity under normal conditions but withdraw liquidity in a systematic and correlated manner under stress, with the single-stock-versus-multi-stock asymmetry documented by Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018) providing perhaps the clearest single piece of evidence for this conditionality. This conditionality has profound implications for market design and systemic risk, and it now generalizes, as the Treasury and sterling episodes demonstrate, well beyond the equity market context in which it was first documented.

The policy response has evolved in two directions. Regulators have imposed circuit breakers—trading halts triggered by price moves beyond specified thresholds—to interrupt feedback loops between algorithmic withdrawal and price decline. The SEC's "limit-up/limit-down" mechanism, introduced following the Flash Crash,

represents an attempt to institutionalize a pause that allows human judgment to re-enter the market during extreme events; the Treasury market has likewise seen the introduction of trading pauses on the interdealer platforms most heavily used by principal trading firms in the wake of the 2014 report. Simultaneously, scholars such as Budish, Cramton, and Shim (2015) have advocated for more fundamental market design changes—particularly the replacement of continuous-time priority queues with frequent batch auctions—to reduce the social waste of the speed arms race while preserving the informational efficiency benefits of algorithmic trading. The physical-infrastructure evidence in Shkilko and Sokolov (2020) suggests that such mechanism-design reforms would need to be paired with restrictions on the private returns to speed investment more generally, since market participants will continue to invest in speed—whether through microwave towers, laser links, or colocation—for as long as the continuous-time limit order book rewards being first.

5. Debate II: Price Discovery Versus Digital Herding and Algorithmic Collusion

5.1 The Efficiency Hypothesis: AI as Information Processor

Beyond liquidity, the second dimension of market quality on which algorithmic trading has provoked sustained academic debate is price discovery—the process by which dispersed private information is aggregated into publicly observable asset prices. The traditional view holds that faster and more sophisticated information processing should accelerate price discovery, and a substantial body of evidence supports this view in the context of algorithmic trading.

The advent of machine learning and natural language processing has dramatically extended the informational reach of algorithmic traders. Large language models can process Federal Reserve statements, earnings call transcripts, sell-side analyst reports, and social media feeds simultaneously, extracting sentiment signals that human traders would require hours or days to synthesize. Lopez de Prado (2018) provides a practitioner's synthesis of how ML methods—including random forests, gradient boosting, recurrent neural networks, and attention-based models—have been deployed to construct predictive signals from both structured (price, volume, order flow) and unstructured (text, images, audio) financial data.

The academic evidence on AI-driven price discovery is concentrated in event-study designs that measure how quickly asset prices incorporate information following a discrete information shock. Brogaard, Hendershott, and Riordan (2014) find that HFTs contribute to permanent price discovery by trading in the direction of efficient price movements—buying on days when prices rise and selling on days when prices fall—and that this contribution is larger for stocks with higher HFT participation. This finding is reinforced by the quote-level decomposition in Brogaard, Hendershott, and Riordan (2019), discussed above in Section 3.1, which shows that a substantial share of the price discovery attributed in earlier studies to HFT *trading* is more precisely attributable to HFT *quoting*—limit orders that are informative about future price movements irrespective of whether they are ultimately executed. Chordia, Roll, and Subrahmanyam (2011) document a secular improvement in short-horizon return predictability across U.S. equity markets over the 1993–2008 period, attributing a significant portion of this improvement to the growth of algorithmic trading and the associated decline in transaction costs.

Cross-country evidence broadens this picture considerably. Boehmer, Fong, and Wu (2021) examine algorithmic trading intensity across 42 exchanges worldwide using co-location and message-traffic data as an instrument for AT activity, and find that algorithmic trading is associated with narrower spreads and improved price efficiency internationally—but simultaneously with an increase in short-term volatility, particularly in periods of market stress. Their identification strategy—exploiting the staggered international rollout of co-location facilities as a source of exogenous variation in AT intensity—provides some of the most convincing causal evidence to date that the liquidity-improving and volatility-increasing effects of algorithmic trading are not mutually exclusive but instead represent two faces of the same underlying phenomenon: faster, more responsive order books that are simultaneously more efficient in normal times and more reactive to shocks. This dual finding directly motivates the herding discussion in Section 5.2. A causally identified domestic counterpart is provided by Li and Zheng (2023), who study a natural experiment—a staggered exchange-level rule change affecting algorithmic order-routing—and similarly find that algorithmic trading causally improves conventional liquidity metrics while raising the sensitivity of prices to order-flow shocks, a pattern consistent with the cross-country evidence in Boehmer, Fong, and Wu (2021).

The competitive dynamics among HFT firms themselves also bear on the efficiency question. Brogaard and Garriott (2019) study the entry of new HFT firms into the Canadian equity market and find that increased competition among HFTs is associated with tighter spreads and improved market quality, but that the marginal benefit of additional HFT entrants declines rapidly—the first few HFT entrants generate the bulk of the liquidity improvement, while subsequent entrants primarily compete away one another's latency-arbitrage rents without generating commensurate additional

social value, a finding that echoes and empirically substantiates the arms-race logic of Budish, Cramton, and Shim (2015).

At the frontier of current research, Ke, Kelly, and Xiu (2023) investigate whether transformer-based language models generate incremental return predictability beyond traditional quantitative signals. Using a large corpus of news articles and a neural-network text-processing pipeline, they construct a "return-predictive" text signal and show that it forecasts subsequent stock returns even after controlling for the sentiment measures used in the pre-transformer NLP literature, and that this predictability is concentrated among stocks with lower analyst coverage and slower institutional ownership turnover—consistent with an interpretation in which the LLM-derived signal captures value-relevant textual information that slower human analysts have not yet fully impounded into prices. This gradual, cross-sectionally heterogeneous incorporation of machine-readable text into prices implies that markets are not yet fully efficient with respect to LLM-extractable textual signals, a conclusion sharpened considerably by more recent work using generative language models directly. Lopez-Lira and Tang (2023) feed raw news headlines into ChatGPT and ask the model simply to classify the sentiment of the headline as good, bad, or unclear for the underlying stock's future price; despite the simplicity of the prompt and the absence of any fine-tuning on financial data, the resulting sentiment scores predict next-day returns with statistically and economically significant magnitude, and a long-short trading strategy built on the model's outputs earns a Sharpe ratio that substantially exceeds that of traditional sentiment-based strategies over their sample period. Because the underlying language model was not trained specifically for financial prediction, this result is difficult to attribute to look-ahead bias from training-data contamination in the way that some early LLM-finance papers have been criticized, and it has accordingly attracted

considerable attention as evidence that general-purpose LLMs possess an emergent capability for financial sentiment extraction that rivals or exceeds purpose-built NLP pipelines. Kirtac and Germano (2024) extend this line of inquiry by systematically comparing sentiment-trading strategies built on OPT, BERT, and FinBERT-family language models against a large news corpus, finding that the best-performing large-model sentiment strategy achieves an annualized Sharpe ratio of approximately 3.05 before transaction costs—a magnitude that, while likely attenuated once realistic trading frictions and capacity constraints are considered, is nonetheless far in excess of the Sharpe ratios documented for classical dictionary-based sentiment measures (e.g., the Loughran-McDonald word lists), suggesting that contextual language understanding, rather than simple keyword counting, is the source of the incremental predictive power.

This finding has implications not only for asset pricing but also for the sociology of market microstructure: if the marginal informed agent is increasingly an AI model rather than a human analyst, the classical assumptions about the nature and distribution of private information may require fundamental revision—a theme we return to in Section 7.3.

5.2 The Herding Hypothesis: When Algorithms Think Alike

The efficiency case for AI-driven price discovery rests implicitly on the assumption that different algorithmic traders are drawing on sufficiently diverse information sets and models to generate heterogeneous—and hence informationally rich—order flow. This assumption is challenged by a growing literature on algorithmic herding: the tendency for correlated algorithmic strategies to generate synchronized order flow that can amplify price movements rather than stabilize them.

The microeconomic mechanism underlying algorithmic herding is straightforward. If a majority of active market participants are using similar machine learning models trained on the same historical data, employing the same momentum or sentiment signals, or responding to the same high-frequency data streams, their strategies will be highly correlated. Under normal conditions, this correlation may be invisible—the strategies generate offsetting order flow from buyers and sellers. But in response to a common signal—a large macro data release, an earnings surprise, or a market-wide liquidity shock—correlated algorithms will fire simultaneously in the same direction, generating a demand shock that overwhelms the available liquidity. The rise of large language models sharpens rather than resolves this concern: because a small number of foundation models (and their fine-tuned derivatives) increasingly underlie a large share of AI-driven trading signals across otherwise-competing firms—precisely the strategies studied by Lopez-Lira and Tang (2023) and Kirtac and Germano (2024)—the diversity of information processing across market participants may in practice be narrowing even as the sophistication of any individual participant's model increases.

O'Hara (2015) and Shleifer and Vishny (1997) together provide theoretical foundations for the herding concern: as trading strategies—whether human arbitrageurs operating with limited capital or algorithmic market-making systems converging on common "best practice" methodologies—become increasingly correlated, the classical limits-to-arbitrage logic implies that a common shock can trigger simultaneous, self-reinforcing withdrawals of capital and liquidity across ostensibly independent participants. Their combined insight predicts that correlated risk management produces fire-sale dynamics: when volatility spikes, similarly-positioned institutions simultaneously reduce their risk exposures, generating the very price declines that

triggered the risk reduction in the first place. This theoretical mechanism has clear empirical counterparts in the behavior of algorithmic traders during the Flash Crash and subsequent stress events, and, as discussed in Section 4.2, in the cross-market Treasury and sterling flash episodes.

The empirical literature on algorithmic herding focuses on two related phenomena: correlated order cancellations and correlated order submissions. Boehmer, Fong, and Wu (2021), discussed above, exploit cross-country variation in co-location rules to estimate the causal effect of HFT on return volatility across 42 stock exchanges. They find that HFT increases short-run volatility—particularly extreme tail volatility—consistent with the herding hypothesis, alongside its liquidity-improving effect documented elsewhere in their analysis. Hirschey (2021), in *Management Science*, provides direct evidence that HFTs engage in "anticipatory trading"—front-running the correlated order flow of other institutions—which can amplify momentum in both directions, a mechanism closely related to the institutional-order front-running documented by Van Kervel and Menkveld (2019) in the Swedish market discussed in Section 4.2. Weller (2018) contributes a further and somewhat distinct concern: using a difference-in-differences design around staggered increases in algorithmic trading intensity, he finds that algorithmic trading is associated with a *reduction* in fundamental information acquisition by human analysts and investors, who rationally substitute away from costly fundamental research toward faster, cheaper reliance on price signals once algorithms have already impounded much of the readily available information into prices. Weller's finding implies a subtle long-run externality of algorithmic trading that is easy to overlook in short-horizon event studies: even if algorithmic trading improves short-run price efficiency, it may erode the very fundamental-information production process on which long-run price discovery ultimately depends, since fewer human

analysts engaged in costly information acquisition means a thinner base of genuinely new information for algorithms themselves to process and disseminate.

The most vivid recent illustration of retail-driven herding, distinct from but related to the institutional algorithmic herding discussed above, is the "meme stock" episode surrounding GameStop Corp. in January 2021, during which a coordinated wave of retail buying—organized substantially through the Reddit forum r/WallStreetBets and executed via commission-free trading applications—drove the stock's price up by several thousand percent within weeks, inflicting substantial losses on hedge funds that had taken large short positions. Lyócsa, Baumöhl, and Výrost (2022) study this episode directly, characterizing it as "YOLO trading"—an acronym for "you only live once" that had become associated with the high-risk, momentum-chasing retail trading style observed on social media during the episode—and documenting that retail order flow and social media sentiment measures jointly predicted the extreme volatility and volume observed in GameStop and a small set of related stocks during the episode, with herding-consistent patterns of retail buying that intensified rather than faded as prices rose, in contrast to the price-contrarian behavior that classical noise-trader models would predict. The GameStop episode is instructive for the broader algorithmic herding debate because it demonstrates that herding dynamics functionally similar to those produced by correlated machine-learning strategies can also emerge from decentralized human coordination mediated by social media and low-friction trading technology—suggesting that the herding concern in modern markets is not reducible to algorithmic trading narrowly defined, but reflects a more general feature of markets in which information and coordination now propagate at internet speed regardless of whether the terminal decision-maker is human or algorithmic.

5.3 Algorithmic Collusion: The Dark Side of Correlated Strategies

A distinct but related concern is algorithmic collusion: the possibility that competing algorithmic traders may sustain supra-competitive outcomes—implicitly colluding on prices or trading strategies—without any explicit communication or agreement between them. The theory of algorithmic collusion has attracted significant attention from both economists and antitrust regulators. Harrington (2018) and Calvano, Calzolari, Denicolò, and Pastorello (2020) demonstrate, using theoretical models and reinforcement-learning simulations respectively, that competing pricing algorithms can learn to sustain collusive outcomes even in repeated game environments with no explicit communication—because the algorithms discover, through trial and error, that unilateral deviations from a coordinated strategy are unprofitable.

In market microstructure, algorithmic collusion could manifest in several ways. Competing HFT market makers might implicitly coordinate on keeping spreads wide in periods of low competition, suppressing the liquidity benefits that are typically attributed to algorithmic market-making—a possibility given some empirical texture by Brogaard and Garriott's (2019) finding that the marginal liquidity benefit of additional HFT competitors declines sharply after the first few entrants, which is consistent with (though not dispositive of) a market structure in which incumbent HFTs face diminishing competitive pressure at the margin. More speculatively, AI-driven traders might develop coordinated strategies for exploiting structural features of electronic limit order books—particularly the order priority rules that govern how matching engines handle simultaneous submissions.

The regulatory implications of algorithmic collusion are still being worked out. Traditional antitrust standards require evidence of explicit agreement or conscious parallelism, neither of which is easily demonstrated when the relevant actors are

competing machine learning systems. Ezrachi and Stucke (2016) argue that existing competition law frameworks are poorly suited to detecting and deterring algorithmic collusion, and that new regulatory tools—including algorithmic auditing requirements and mandatory disclosure of strategy parameters—may be necessary.

6. Debate III: Algorithmic Manipulation and the Limits of Regulatory Penetration

6.1 New Manipulative Tactics: Spoofing and Quote Stuffing

The third dimension of market quality addressed in this review is market integrity—the extent to which trading activity reflects genuine supply and demand rather than manipulation. The ascent of algorithmic trading has not merely changed the speed and scale of legitimate market activity; it has also generated new forms of market manipulation that exploit the specific features of electronic limit order books and the information asymmetries created by speed differentials between participants.

The most extensively documented form of algorithmic manipulation is spoofing: the submission of large limit orders on one side of the book with the intent to cancel them before execution, thereby creating a misleading impression of supply or demand that induces other traders to transact at unfavorable prices. The spoofer profits by trading in the direction opposite to the artificial signal before cancelling the deceptive orders. Spoofing is illegal under the Dodd-Frank Act (2010) in the United States and under equivalent provisions in European and Asian financial regulation, but its detection and prosecution remain challenging because the activity is indistinguishable in aggregate statistics from legitimate order management.

Comerton-Forde and Putniņš (2015) develop an econometric methodology for detecting spoofing and, more broadly, stock price manipulation in order-book and transaction data, identifying episodes in which the submission and near-instantaneous cancellation of large orders are associated with transient price movements and subsequent reversal—the expected footprint of a successful spoof. Their broader contribution documents that manipulation is far from a rare tail phenomenon: using regulatory enforcement data matched to market characteristics, they estimate the prevalence of manipulation across the market and identify the stock- and market-level characteristics—including low liquidity, high retail participation, and weak analyst coverage—that predict where manipulation is more likely to occur, providing an empirical base rate against which the AI-facilitated manipulation concerns discussed below should be calibrated. Their methodology has been adopted, with modifications, by securities regulators in multiple jurisdictions. A prominent enforcement case is the prosecution of Navinder Sarao, whose algorithmic spoofing strategy in the E-mini S&P 500 futures market on May 6, 2010 was subsequently identified as a contributing factor to the Flash Crash—though Kirilenko et al. (2017) suggest that Sarao's contribution, while real, was smaller than initial accounts suggested.

Quote stuffing—the deliberate submission of massive order volumes to degrade the information quality available to competing algorithms—is more difficult to prove as intentional manipulation, because it is often indistinguishable from legitimate high-frequency market-making or arbitrage. Egginton, Van Ness, and Van Ness (2016) provide systematic evidence that quote-stuffing episodes are associated with increased spreads and reduced depth for affected stocks, consistent with an intentional information-disruption strategy. Whether quote stuffing constitutes manipulation in a

legal sense depends on the intent behind the order submission, which regulators have found difficult to establish beyond reasonable doubt.

A related, though conceptually distinct, threat to market integrity arising from market fragmentation is the growth of dark pools—trading venues that do not display pre-trade quotes and that have grown substantially alongside the rise of algorithmic and high-frequency trading, partly as a defensive response by institutional investors seeking to avoid the very front-running behavior documented by Van Kervel and Menkveld (2019) in lit markets. Zhu (2014) provides the canonical theoretical and empirical treatment of this phenomenon, modeling informed and uninformed traders' venue choice between a lit exchange and a dark pool and showing that dark pools tend to attract a disproportionate share of uninformed order flow (because informed traders, who benefit from immediate execution and are more sensitive to non-execution risk, prefer the lit market), which paradoxically *concentrates* adverse selection risk on the lit exchange and can, beyond a threshold level of dark-pool trading volume, impair rather than improve overall price discovery. Zhu's result is significant for the algorithmic manipulation debate because it establishes that fragmentation across lit and dark venues—itself substantially a product of algorithmic order-routing technology—creates structural adverse-selection dynamics that are difficult to distinguish from, and may in practice be compounded by, deliberate manipulative order-routing strategies. Complementary evidence on fragmentation comes from Bernales, Garrido, Sagade, Valenzuela, and Westheide (2020), who model and empirically examine the trade-off HFT liquidity suppliers face between the benefits of liquidity provision across fragmented markets and the "picking-off" risk of being adversely selected by faster or better-informed competitors, showing that this trade-off shapes the cross-venue distribution of HFT quoting activity in ways that have direct implications for the

effectiveness of any single venue's surveillance regime, since manipulative or exploitative strategies can be structured to straddle multiple fragmented venues precisely to evade single-venue detection.

6.2 Regulatory Responses: Weaponizing AI for Market Surveillance

The regulatory response to algorithmic manipulation has, with some irony, increasingly employed the same AI tools that the manipulators themselves use. The SEC's Office of Analytics and Research has developed AI-powered surveillance systems—including the Consolidated Audit Trail (CAT), which captures order-level data across all U.S. exchanges—that can identify anomalous patterns of order submission and cancellation in real time. ESMA, the European Securities and Markets Authority, has similarly developed algorithmic trading surveillance frameworks under MiFID II that require trading firms to test and document their algorithms' behavior under stress scenarios.

The SEC Office of Analytics and Research (2014) literature review synthesizes the state of AI- and rule-based manipulation-detection approaches available to regulators at the time, cataloguing the order-level statistics—message-to-trade ratios, cancellation timing, and order-book imbalance patterns—that supervised classifiers use to flag potential spoofing episodes. The review's assessment is that machine learning classifiers trained on labeled examples of confirmed manipulation can achieve materially higher detection accuracy than static rule-based thresholds, reducing both false positives (which impose compliance costs on legitimate traders) and false negatives (which allow manipulation to go undetected), though the review also cautions that classifier performance is highly sensitive to the representativeness of the labeled

training sample—consistent with the intuition that spoofers create a distinctive "fingerprint" in order-book data that adapts as detection methods improve.

The limits of algorithmic surveillance are, however, significant. First, supervised machine learning requires labeled training data—confirmed cases of manipulation—which are scarce and potentially unrepresentative of future manipulative strategies. As regulators train their detection models on historical manipulation cases, sophisticated manipulators adapt their strategies to evade the specific features that surveillance models are trained to detect. This adversarial dynamic—the regulatory analogue of the speed arms race quantified by Aquilina, Budish, and O'Neill (2022) and Shkilko and Sokolov (2020)—implies that manipulation detection is inherently a moving target. Second, the volume of order data generated by modern algorithmic markets is so large that even AI-powered surveillance systems face fundamental challenges of scalability and latency. A spoofing strategy that operates at millisecond timescales may execute and reverse before a surveillance system processing thousands of messages per second can flag it. This scalability challenge is compounded by market fragmentation of the kind studied by Bernales, Garrido, Sagade, Valenzuela, and Westheide (2020): a strategy that appears innocuous within any single venue's audit trail may be manipulative when the trader's full, cross-venue order-submission pattern is considered, yet most surveillance systems remain organized around single-venue data feeds.

Third, and most conceptually challenging, the distinction between manipulation and legitimate market-making is inherently blurry. A market maker who cancels limit orders in response to adverse information is engaging in legitimate order management; a spoofer who cancels limit orders to extract profits from the price impact of the cancellation is engaging in manipulation. The behavioral difference is one of intent—

and intent is not directly observable by a surveillance algorithm, which can only infer it from the pattern of order activity. The legal and regulatory framework for algorithmic manipulation thus faces a fundamental inference problem that cannot be resolved by purely technical means. The causal evidence in Li and Zheng (2023) is instructive here: because their natural-experiment design isolates the causal effect of a specific, exogenously imposed algorithmic-trading rule change on market quality, it offers regulators a template for the kind of quasi-experimental evaluation that would be needed to distinguish the beneficial market-quality effects of legitimate algorithmic strategies from the harmful effects of manipulative ones—an evaluation that purely observational, classifier-based surveillance of the kind reviewed by the SEC Office of Analytics and Research (2014) cannot, on its own, deliver.

6.3 Natural Language Processing in Regulatory Enforcement

A final frontier in regulatory AI is the application of natural language processing to the supervision of algorithmic traders' communications and disclosures. Under MiFID II in Europe and equivalent SEC rules in the United States, firms employing algorithmic trading strategies are required to maintain detailed records of their algorithms' logic, testing procedures, and risk controls. These records are voluminous, highly technical, and previously reviewed by human compliance officers at considerable cost and delay.

Recent work has explored the use of NLP models to automate the review of algorithmic trading disclosures, identifying potential compliance failures or risk control gaps in the submitted documentation. While systematic academic evidence on NLP-based review of HFT strategy filings specifically remains at an early stage, the conceptual potential is clear: regulators equipped with the same language models that

algorithmic traders use to analyze market information—including the general-purpose large language models studied by Lopez-Lira and Tang (2023) and the domain-tuned models studied by Kirtac and Germano (2024)—could substantially reduce the information asymmetry between supervisor and supervised. The dual-use character of this technology is a theme to which we return in the synthesis that follows: the very language-model capabilities that allow a trading firm to extract alpha from an earnings call transcript in milliseconds are, in principle, equally available to the regulator seeking to parse that firm's own algorithmic risk-control documentation.

7. Synthesis: Toward a Microstructure Theory of Algorithmic Agents

7.1 The Paradigm Shift

The three debates reviewed in Sections 4 through 6 share a common deeper structure: they all reflect the inadequacy of classical market microstructure theory—built on the assumption of rational human agents with diverse information sets and limited information-processing capacity—for understanding markets dominated by algorithmic agents. The synthesis proposed here is not merely that human traders have been replaced by machines, but that this replacement constitutes a qualitative shift in the nature of the actors that populate market microstructure models, with implications that classical theory cannot accommodate.

In the canonical Kyle (1985) model, the insider's trading strategy is bounded by the constraint that she can only act on private information, not react to the strategies of other agents in real time. In an algorithmic market, this assumption is untenable: competing algorithms continuously observe each other's order flow and adapt their strategies instantaneously. The game-theoretic structure of algorithmic market

interaction is therefore closer to a multi-agent reinforcement learning problem than to the sequential equilibria of classical microstructure theory. This observation has motivated a growing literature on "agent-based models" of electronic markets, in which the emergent properties of market microstructure—including flash crashes, liquidity spirals, and herding dynamics—arise from the interaction of simple adaptive agents rather than from equilibrium conditions derived analytically (Farmer and Foley, 2009; Paddrik, Hayes, Todd, and Yang, 2012). The finding that limit orders rather than trades are the primary channel of price discovery (Brogaard, Hendershott, and Riordan, 2019) reinforces this reframing: a theory adequate to modern markets must characterize the informational content of the entire, continuously updated limit order book—including cancelled and unexecuted orders—rather than the sequence of realized trades alone.

The market-design implications of this paradigm shift have themselves become an active theoretical research frontier. Menkveld and Zoican (2017) model exchange latency directly, showing that reducing the exchange's own processing latency does not uniformly improve liquidity: faster exchanges can worsen adverse selection by giving high-frequency market makers a shorter window in which to update stale quotes ahead of informed order flow, so that the relationship between speed and market quality is non-monotonic rather than the simple "faster is better" intuition that motivated early colocation investment. Pagnotta and Philippon (2018) develop a complementary equilibrium framework in which trading venues compete on speed as a product-differentiation choice, showing that the socially optimal amount of speed investment depends on the elasticity of investor demand for fast trading and need not coincide with the equilibrium level that competing exchanges actually deliver—formalizing, in a market-design setting, the same wasteful-arms-race intuition that motivates Budish, Cramton, and Shim's (2015) frequent batch auction proposal. Together, these

contributions suggest that the speed arms race documented empirically throughout Sections 4 through 6 is not merely a byproduct of technology but an equilibrium outcome of exchange and market-maker competition that can, in principle, be redirected through deliberate market design.

7.2 The Correlated Failure Mode

A second key insight from the survey is that the risk profile of algorithmic markets is qualitatively different from that of human-dominated markets in a way that classical theory did not anticipate. In a market populated by heterogeneous human traders with diverse risk preferences, time horizons, and information sources, the aggregate effect of individual failures is diversified: when one trader is distressed and must liquidate, other traders are available to absorb the supply. In a market dominated by a small number of highly correlated algorithmic strategies—all using similar risk management systems, responding to the same data feeds, subject to similar regulatory constraints—individual failures are correlated by construction.

This correlated failure mode is the microstructure expression of the systemic risk concerns that have motivated macroprudential regulation since the Global Financial Crisis. The Flash Crash is the most dramatic illustration, but the phenomenon recurs in less severe form across a wide range of electronic markets: in the Treasury flash crash of October 2014, documented in the joint staff report of the U.S. Department of the Treasury et al. (2015); in the sterling flash event of October 2016, documented by the BIS Markets Committee (2017); in the multi-stock extreme-price-movement episodes studied by Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018); and in the equity volatility explosion of February 2018, when a family of short-volatility ETPs that had effectively become algorithmic market makers in the VIX

futures market imploded simultaneously. The GameStop episode of January 2021, discussed in Section 5.2, provides a further and instructive variant: there, the correlated "failure" ran in the opposite direction, with correlated *buying* pressure among retail traders inflicting losses on short-sellers rather than the more familiar pattern of correlated selling inflicting losses on long positions—demonstrating that the correlated-agent fragility identified throughout this review is a general structural feature of internet-speed, socially or algorithmically coordinated markets rather than a phenomenon confined to any particular direction of price movement or class of trading technology. In each case, the correlation of strategies—whether algorithmic, retail-social, or some hybrid of the two—rather than the failure of any single market participant, was the proximate driver of extreme market dysfunction.

The cross-country evidence in Boehmer, Fong, and Wu (2021) and the natural-experiment evidence in Li and Zheng (2023) together suggest that this correlated-fragility tendency is not an artifact of any single market's institutional design but a general consequence of algorithmic trading intensity, observable across markets with quite different regulatory architectures, trading protocols, and participant compositions. This generality is precisely what one would expect if the underlying driver is the correlated-strategy, limits-to-arbitrage mechanism proposed by O'Hara (2015) and Shleifer and Vishny (1997) rather than any institution-specific feature of U.S. market structure.

7.3 The Informational Arms Race and Its Discontents

The third synthetic observation concerns the changing nature of the informational foundations of price discovery. Classical theory posits that prices incorporate information because informed agents—those with private knowledge of

asset fundamentals—trade in ways that reveal their information to market makers, gradually driving prices toward true value. In AI-enhanced markets, the "information" that drives price movements is increasingly short-lived, statistical, and derived from the processing of publicly observable data rather than genuinely private signals. The LLM that reads a central bank statement, or a corporate earnings headline, faster than human analysts is not informed in the classical sense—its informational advantage is purely technological, a distinction that the results of Lopez-Lira and Tang (2023) and Kirtac and Germano (2024) throw into particularly sharp relief, since the underlying "information" in both cases is a public news headline that any human trader could, in principle, read; what the LLM contributes is not access to new information but the speed and consistency of interpreting old information at a scale no team of human analysts could match.

This distinction matters for welfare. If the social value of price discovery derives from the aggregation of genuinely private information about economic fundamentals—valuations, growth prospects, technology—then the speed-driven "price discovery" of algorithmic traders may deliver smaller welfare benefits than classical theory would predict. Conversely, if the socially relevant information is precisely the kind of high-frequency, publicly observable signal that AI excels at processing—order flow imbalances, sentiment shocks, cross-asset correlations—then the case for algorithmic price discovery is stronger. Weller's (2018) finding that algorithmic trading may crowd out human fundamental-information acquisition suggests a further complication: even the supply of genuinely private, fundamentals-based information may itself be endogenous to the growth of algorithmic trading, shrinking over time as human analysts rationally substitute toward faster, cheaper reliance on algorithmically generated price signals. Resolving this question requires a clearer understanding of what information

asset prices are supposed to aggregate, and for whom—a question that has become both empirically and normatively pressing as AI has taken over the role of marginal informed trader.

8. Conclusion

This paper has reviewed a broad and rapidly expanding literature on the microstructural consequences of algorithmic trading, high-frequency trading, and artificial intelligence in financial markets, synthesizing 60 studies identified through a systematic, PRISMA-guided review of 847 initial records across four databases. Three central conclusions emerge.

First, the relationship between algorithmic trading and market quality is genuinely conditional: algorithms improve measured liquidity and price efficiency under normal conditions but introduce new failure modes—phantom liquidity, correlated withdrawal, and herding-induced volatility spirals—that may be more damaging to market quality than the frictions they displace. This conditionality, first documented in the equity context by Kirilenko, Kyle, Samadi, and Tuzun (2017) and sharpened considerably by the single-stock-versus-multi-stock distinction in Brogaard, Carrion, Moyaert, Riordan, Shkilko, and Sokolov (2018), now has a well-established cross-asset evidentiary base spanning U.S. equities, U.S. Treasuries, and foreign exchange, and a well-established cross-country evidentiary base spanning at least 42 international exchanges (Boehmer, Fong, and Wu, 2021). The policy implications are that regulatory frameworks must be calibrated to the conditional nature of algorithmic liquidity, with circuit breakers and market design reforms targeted specifically at the tail-event behavior of algorithmic traders, and with quantitative attention to the arms-

race costs documented by Aquilina, Budish, and O'Neill (2022) and Shkilko and Sokolov (2020).

Second, the dual-use character of AI in financial markets—simultaneously improving informational efficiency and potentially enabling algorithmic collusion, herding, and manipulation—poses fundamental challenges for both theory and regulation. The same large language model capabilities that allow Lopez-Lira and Tang (2023) and Kirtac and Germano (2024) to demonstrate genuinely novel return predictability from public text are, in principle, available to every market participant simultaneously, raising the herding concern developed in Section 5.2 and given vivid, non-algorithmic illustration by the GameStop episode studied by Lyócsa, Baumöhl, and Výrost (2022). The same surveillance-relevant pattern-recognition capabilities that allow regulators to detect spoofing and evaluate risk disclosures (SEC Office of Analytics and Research, 2014) are constrained by the same adversarial dynamics, market fragmentation (Zhu, 2014; Bernales, Garrido, Sagade, Valenzuela, and Westheide, 2020), and intent-inference problems that have always limited manipulation enforcement. Regulatory frameworks must grapple with this duality by developing adaptive, AI-augmented surveillance tools while simultaneously maintaining the legal flexibility to address manipulation tactics that existing frameworks do not contemplate.

Third, and most fundamentally, the dominance of algorithmic agents in contemporary financial markets requires a paradigm shift in market microstructure theory. The rational human agent of classical microstructure must yield to the algorithmic agent of contemporary reality—an agent characterized not by private fundamental information but by computational speed, adaptive learning, correlated strategies, and emergent behaviors that classical equilibrium models are ill-equipped to capture. The most promising theoretical directions involve agent-based simulation,

multi-agent reinforcement learning, and network-theoretic models of contagion, extended to account for the finding that price discovery operates substantially through quoting rather than trading (Brogaard, Hendershott, and Riordan, 2019) and for the possibility that fundamental information acquisition itself is endogenous to the growth of algorithmic trading (Weller, 2018). Building these frameworks, and testing them against the rich empirical record that modern electronic markets generate—a record that now spans equities, Treasuries, foreign exchange, and the retail-social hybrid markets exemplified by the GameStop episode—is the most pressing research agenda in financial microstructure today.

Declaration of Generative AI Use

Generative AI tools were used to assist with aspects of literature synthesis and manuscript drafting. All factual claims, interpretations, and conclusions remain the sole intellectual responsibility of the authors, who have reviewed and verified the final content in its entirety.

References

1. Ait-Sahalia, Y., & Saglam, M. (2013). High frequency traders: Taking advantage of speed. *NBER Working Paper No. 19531*. <https://doi.org/10.3386/w19531>
2. Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
[https://doi.org/10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6)

3. Aquilina, M., Budish, E. B., & O'Neill, P. (2022). Quantifying the high-frequency trading "arms race." *Quarterly Journal of Economics*, 137(1), 493–564.
<https://doi.org/10.1093/qje/qjab032>
4. Bank for International Settlements, Markets Committee. (2011). *High-frequency trading in the foreign exchange market*. BIS Markets Committee Publication No. 5. <https://www.bis.org/publ/mktc05.pdf>
5. Baron, M., Brogaard, J., Hagströmer, B., & Kirilenko, A. (2019). Risk and return in high-frequency trading. *Journal of Financial and Quantitative Analysis*, 54(3), 993–1024. <https://doi.org/10.1017/S0022109018000725>
6. Bernales, A., Garrido, N., Sagade, S., Valenzuela, M., & Westheide, C. (2020). Trader competition in fragmented markets: Liquidity supply versus picking off risk. *Journal of Financial and Quantitative Analysis*, 55(2), 542–579.
<https://doi.org/10.1017/S0022109018001584>
7. BIS Markets Committee. (2017). *The sterling "flash event" of 7 October 2016*. Bank for International Settlements. <https://www.bis.org/publ/mktc09.pdf>
8. Boehmer, E., Fong, K. Y. L., & Wu, J. (2021). Algorithmic trading and market quality: International evidence. *Journal of Financial and Quantitative Analysis*, 56(8), 2659–2688. <https://doi.org/10.1017/S0022109021000031>
9. Brogaard, J., Carrion, A., Moyaert, T., Riordan, R., Shkilko, A., & Sokolov, K. (2018). High frequency trading and extreme price movements. *Journal of Financial Economics*, 128(2), 253–265.
<https://doi.org/10.1016/j.jfineco.2018.02.002>

10. Brogaard, J., & Garriott, C. (2019). High-frequency trading competition. *Journal of Financial and Quantitative Analysis*, 54(4), 1469–1497.
<https://doi.org/10.1017/S0022109018001225>
11. Brogaard, J., Hagströmer, B., Nordén, L., & Riordan, R. (2015). Trading fast and slow: Colocation and liquidity. *Review of Financial Studies*, 28(12), 3407–3443. <https://doi.org/10.1093/rfs/hhv032>
12. Brogaard, J., Hendershott, T., & Riordan, R. (2014). High-frequency trading and price discovery. *Review of Financial Studies*, 27(8), 2267–2306.
<https://doi.org/10.1093/rfs/hhu032>
13. Brogaard, J., Hendershott, T., & Riordan, R. (2019). Price discovery without trading: Evidence from limit orders. *Journal of Finance*, 74(4), 1621–1658.
<https://doi.org/10.1111/jofi.12769>
14. Brunnermeier, M. K., & Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.
<https://doi.org/10.1093/rfs/hhn098>
15. Budish, E., Cramton, P., & Shim, J. (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *Quarterly Journal of Economics*, 130(4), 1547–1621. <https://doi.org/10.1093/qje/qjv027>
16. Budish, E., Lee, R. S., & Shim, J. (2024). A theory of stock exchange competition and innovation: Will the market fix the market? *Journal of Political Economy*, 132(4), 1209–1269. <https://doi.org/10.1086/727284>
17. Calvano, E., Calzolari, G., Denicolò, V., & Pastorello, S. (2020). Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10), 3267–3297. <https://doi.org/10.1257/aer.20190623>

18. Carrion, A. (2013). Very fast money: High-frequency trading on the NASDAQ. *Journal of Financial Markets*, 16(4), 680–711.
<https://doi.org/10.1016/j.finmar.2013.06.005>
19. CFTC and SEC. (2010). *Findings regarding the market events of May 6, 2010*. Report of the staffs of the CFTC and SEC to the Joint Advisory Committee on Emerging Regulatory Issues.
20. Chaboud, A. P., Chiquoine, B., Hjalmarsson, E., & Vega, C. (2014). Rise of the machines: Algorithmic trading in the foreign exchange market. *Journal of Finance*, 69(5), 2045–2084. <https://doi.org/10.1111/jofi.12186>
21. Chordia, T., Roll, R., & Subrahmanyam, A. (2011). Recent trends in trading activity and market quality. *Journal of Financial Economics*, 101(2), 243–263.
<https://doi.org/10.1016/j.jfineco.2011.03.008>
22. Comerton-Forde, C., & Putniņš, T. J. (2015). Stock price manipulation: Prevalence and determinants. *Review of Finance*, 19(4), 1551–1588.
<https://doi.org/10.1093/rof/rfu023>
23. Egginton, J. F., Van Ness, B. F., & Van Ness, R. A. (2016). Quote stuffing. *Financial Management*, 45(3), 583–608. <https://doi.org/10.1111/fima.12126>
24. Ezrachi, A., & Stucke, M. E. (2016). *Virtual competition: The promise and perils of the algorithm-driven economy*. Harvard University Press.
25. Farmer, J. D., & Foley, D. (2009). The economy needs agent-based modelling. *Nature*, 460(7256), 685–686. <https://doi.org/10.1038/460685a>
26. Foucault, T., Hombert, J., & Roşu, I. (2016). News trading and speed. *Journal of Finance*, 71(1), 335–382. <https://doi.org/10.1111/jofi.12302>

27. Gai, J., Yao, C., & Ye, M. (2012). The externalities of high frequency trading.
Working Paper, University of Illinois at Urbana-Champaign.
<https://ssrn.com/abstract=2066839>
28. Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100. [https://doi.org/10.1016/0304-405X\(85\)90044-3](https://doi.org/10.1016/0304-405X(85)90044-3)
29. Goldstein, M. A., Kumar, P., & Graves, F. C. (2014). Computerized and high-frequency trading. *Financial Review*, 49(2), 177–202.
<https://doi.org/10.1111/fire.12031>
30. Harrington, J. E. (2018). Developing competition law for collusion by autonomous artificial agents. *Journal of Competition Law & Economics*, 14(3), 331–363. <https://doi.org/10.1093/joclec/nhy016>
31. Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance*, 46(1), 179–207. <https://doi.org/10.1111/j.1540-6261.1991.tb03749.x>
32. Hautsch, N., & Horvath, A. (2019). How large is liquidity risk in an automated, electronic market? *Journal of Financial Markets*, 45, 59–79.
<https://doi.org/10.1016/j.finmar.2019.03.001>
33. Hendershott, T., Jones, C. M., & Menkveld, A. J. (2011). Does algorithmic trading improve liquidity? *Journal of Finance*, 66(1), 1–33.
<https://doi.org/10.1111/j.1540-6261.2010.01624.x>
34. Hendershott, T., & Riordan, R. (2013). Algorithmic trading and the market for liquidity. *Journal of Financial and Quantitative Analysis*, 48(4), 1001–1024.
<https://doi.org/10.1017/S0022109013000471>

35. Hirschey, N. (2021). Do high-frequency traders anticipate buying and selling pressure? *Management Science*, 67(6), 3321–3345.
<https://doi.org/10.1287/mnsc.2020.3608>
36. Jones, C. M. (2013). What do we know about high-frequency trading? *Columbia Business School Research Paper No. 13-11*.
<https://ssrn.com/abstract=2236201>
37. Jovanovic, B., & Menkveld, A. J. (2016). Middlemen in limit order markets. *Working Paper*, SSRN. <https://ssrn.com/abstract=1624329>
38. Ke, Z. T., Kelly, B. T., & Xiu, D. (2023). Predicting returns with text data. *Journal of Financial Economics*, 149(2), 422–453.
<https://doi.org/10.1016/j.jfineco.2023.05.003>
39. Kirilenko, A., Kyle, A. S., Samadi, M., & Tuzun, T. (2017). The flash crash: High-frequency trading in an electronic market. *Journal of Finance*, 72(3), 967–998. <https://doi.org/10.1111/jofi.12498>
40. Kirtac, K., & Germano, G. (2024). Sentiment trading with large language models. *Finance Research Letters*, 62, article 105182.
<https://doi.org/10.1016/j.frl.2024.105182>
41. Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335. <https://doi.org/10.2307/1913210>
42. Li, J., & Zheng, L. (2023). The causal impact of algorithmic trading on market quality: Evidence from a natural experiment. *Journal of Financial and Quantitative Analysis*, 58(1), 155–182.
<https://doi.org/10.1017/S0022109022000883>
43. Lopez de Prado, M. (2018). *Advances in financial machine learning*. Wiley.

44. Lopez-Lira, A., & Tang, Y. (2023). Can ChatGPT forecast stock price movements? Return predictability and large language models. *Review of Financial Studies*, 37(6), 1897–1928. <https://doi.org/10.1093/rfs/hhae066>
45. Lyócsa, Š., Baumöhl, E., & Výrost, T. (2022). YOLO trading: Riding with the herd during the GameStop episode. *Finance Research Letters*, 46, article 102359. <https://doi.org/10.1016/j.frl.2021.102359>
46. Menkveld, A. J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets*, 16(4), 712–740. <https://doi.org/10.1016/j.finmar.2013.06.006>
47. Menkveld, A. J., & Yueshen, B. Z. (2019). The flash crash: A cautionary tale about highly fragmented markets. *Management Science*, 65(10), 4470–4488. <https://doi.org/10.1287/mnsc.2018.3040>
48. Menkveld, A. J., & Zoican, M. A. (2017). Need for speed? Exchange latency and liquidity. *Review of Financial Studies*, 30(4), 1188–1228. <https://doi.org/10.1093/rfs/hhx006>
49. O'Hara, M. (2015). High frequency market microstructure. *Journal of Financial Economics*, 116(2), 257–270. <https://doi.org/10.1016/j.jfineco.2015.01.003>
50. Paddrik, M., Hayes, R., Todd, A., & Yang, S. (2012). *An agent-based model of the E-mini S&P 500 applied to flash crash analysis*. CFTC Working Paper.
51. Pagnotta, E., & Philippon, T. (2018). Competing on speed. *Econometrica*, 86(3), 1067–1115. <https://doi.org/10.3982/ECTA10762>
52. SEC Office of Analytics and Research. (2014). *Equity market structure literature review, Part II: High frequency trading*. U.S. Securities and Exchange

Commission.

https://www.sec.gov/marketstructure/research/hft_lit_review_march_2014.pdf

53. Shkilko, A., & Sokolov, K. (2020). Every cloud has a silver lining: Fast trading, microwave connectivity, and trading costs. *Journal of Finance*, 75(6), 2899–2940. <https://doi.org/10.1111/jofi.12969>
54. Shleifer, A., & Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1), 35–55. <https://doi.org/10.1111/j.1540-6261.1997.tb03807.x>
55. U.S. Department of the Treasury, Board of Governors of the Federal Reserve System, Federal Reserve Bank of New York, U.S. Securities and Exchange Commission, & U.S. Commodity Futures Trading Commission. (2015). *Joint staff report: The U.S. Treasury market on October 15, 2014*. <https://home.treasury.gov/system/files/276/joint-staff-report-the-us-treasury-market-on-10-15-2014.pdf>
56. Van Kervel, V., & Menkveld, A. J. (2019). High-frequency trading around large institutional orders. *Journal of Finance*, 74(3), 1091–1137. <https://doi.org/10.1111/jofi.12759>
57. Weller, B. M. (2018). Does algorithmic trading reduce information acquisition? *Review of Financial Studies*, 31(6), 2184–2226. <https://doi.org/10.1093/rfs/hhx137>
58. Zhang, F. (2010). High-frequency trading, stock volatility, and price discovery. *SSRN Working Paper*. <https://ssrn.com/abstract=1691679>
59. Zhou, H., Kalev, P. S., & Frino, A. (2020). Algorithmic trading in turbulent markets. *Pacific-Basin Finance Journal*, 62, article 101374. <https://doi.org/10.1016/j.pacfin.2020.101374>

60. Zhu, H. (2014). Do dark pools harm price discovery? *Review of Financial Studies*, 27(3), 747–789. <https://doi.org/10.1093/rfs/hht078>